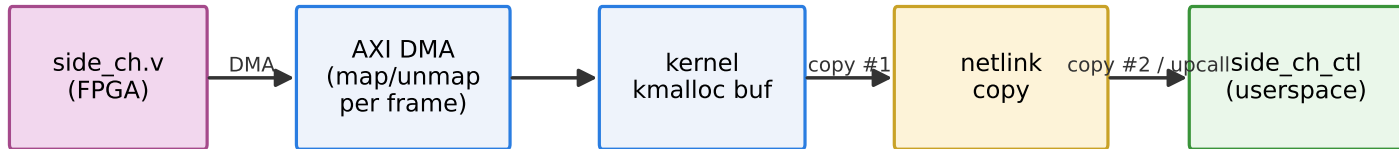
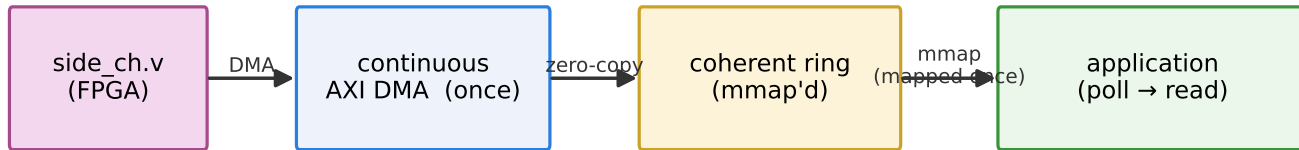


(a) Stock openwifi CSI path



poll every ~1 ms · CPU copy ×2 · per-frame map/unmap

(b) Proposed zero-copy CSI path



1× DMA · 0× CPU copy · interrupt-driven